



اونيورسيتي مليسيا قهغ السلطان عبد الله
UNIVERSITI MALAYSIA PAHANG
AL-SULTAN ABDULLAH

FACULTY OF COMPUTING

BCS3263

SOFTWARE QUALITY ASSURANCE PLAN

EDUQUEST

MANAGE ANNOUNCEMENT AND DISCUSSION

SECTION 01B

LECTURER: ROSLINA BINTI SIDEK

NAME	STUDENT ID
NUR ALYA SYAKIRAH BINTI NASARUDIN	CB22141

TABLE OF CONTENT

1.0 PURPOSE AND SCOPE	1
1.1 PURPOSE	1
1.2 SCOPE	2
2.0 DEFINITIONS AND ACRONYMS	3
2.1 DEFINITIONS	3
2.2 ACRONYMS	4
3.0 REFERENCES DOCUMENTS	5
4.0 SQA PLAN OVERVIEW	6
4.1 ORGANIZATION AND INDEPENDENCE	6
4.2 SOFTWARE PRODUCT RISK	10
4.3 TOOLS	11
4.4 STANDARD, PRACTICES AND CONVENTIONS	12
4.5 EFFORT, RESOURCES AND SCHEDULE	13
4.5.1 EFFORT	13
4.5.2 RESOURCES	15
4.5.3 SCHEDULE	16
5.0 ACTIVITIES, OUTCOMES AND TASKS	17
5.1 PRODUCT ASSURANCE	17
5.1.1 FUNCTIONALITY	17
5.1.2 PERFORMANCE	20
5.1.3 USABILITY	22
5.2 PROCESS ASSURANCE	24
6.0 ADDITIONAL CONSIDERATIONS	26
6.1 CONTRACT REVIEW	26
6.2 QUALITY MEASUREMENT	26
6.3 WAIVERS AND DEVIATIONS	27
6.4 TASK REPETITION	27
6.5 RISK TO PERFORMING SQA	27
6.6 COMMUNICATIONS STRATEGY	28
6.7 NON-CONFORMANCE PROCESS	28
7.0 SQA RECORDS	29
7.1 ANALYZE, IDENTIFY, COLLECT, FILE, MAINTAIN AND DISPOSE	29
7.2 AVAILABILITY OF RECORDS	29

1.0 PURPOSE AND SCOPE

1.1 PURPOSE

The purpose of this Software Quality Assurance Plan (SQAP) is to ensure that the Manage Announcement and Discussion Module of the EduQuest Teaching and Learning Management System (TLMS) meets the required quality standards as defined in the IEEE 730-2014 guidelines for Software Quality Assurance. This plan defines the framework, methodologies and procedures necessary to verify that the module's design, implementation and performance adhere to both functional and non-functional quality requirements.

The Manage Announcement and Discussion module plays a critical role in enhancing communication and collaboration among all users within the EduQuest platform including administrators, lecturers and students. It serves as a centralized platform where lecturers can create, edit and delete important announcements, while students can view updates, participate in academic discussions and interact with their peers and lecturers. Administrators oversee the content to ensure that communication remains appropriate and aligned with institutional guidelines.

This SQAP establishes the quality control mechanisms that will be applied to the module throughout the development lifecycle from requirement validation, design review, coding and testing, to final acceptance and maintenance [1]. It ensures that all implemented features including posting announcements, replying to discussion threads, searching and filtering topics, and content moderation function accurately and efficiently. The plan also aims to minimize defects, enhance user experience and ensure data integrity across all module operations.

Furthermore, this SQAP emphasizes adherence to defined standards and systematic testing practices to confirm that the software performs reliably under different usage conditions. The ultimate goal of this plan is to guarantee that the Manage Announcement and Discussion module provides a user-friendly and interactive communication environment, fostering engagement, collaboration and effective information exchange between educators and learners in the EduQuest platform [2].

1.2 SCOPE

This Software Quality Assurance Plan (SQAP) applies specifically to the Manage Announcement and Discussion Module of the EduQuest system [3]. It covers all functionalities that support the posting, viewing, editing, searching and deletion of announcements and discussion threads. The plan focuses on ensuring that each function performs as intended without system errors, data loss, or unauthorized access.

The scope also includes quality assurance activities such as requirement validation, functional testing, performance evaluation, and usability assessment. This SQAP will guide all the testing and validation process throughout the development lifecycle from design and implementation to integration, deployment and maintenance to maintain overall system quality and user satisfaction.

2.0 DEFINITIONS AND ACRONYMS

2.1 DEFINITIONS

Software Quality Assurance Plan (SQAP) :

A Software Quality Assurance Plan is a formal document that describes the policies, procedures, tools and responsibilities necessary to achieve and maintain software quality throughout the project lifecycle. It outlines the specific quality objectives, testing strategies, standards and verification methods that will be implemented to ensure the developed system meets its functional and non-functional requirements.

EduQuest System :

EduQuest is a web-based Teaching and Learning Management System (TLMS) designed to streamline educational operations and enhance communication between administrators, lecturers and students. The system includes multiple modules such as manage profile, manage assignment and notes, and manage announcement and discussion where each is responsible for specific educational functions.

Manage Announcement and Discussion Module :

The Manage Announcement and Discussion Module is responsible for enabling lecturers to post, edit and delete announcements while allowing students to view updates, participate in discussions and reply to threads. Administrators monitor content to maintain appropriate communication and ensure a collaborative and organized learning environment.

2.2 ACRONYMS

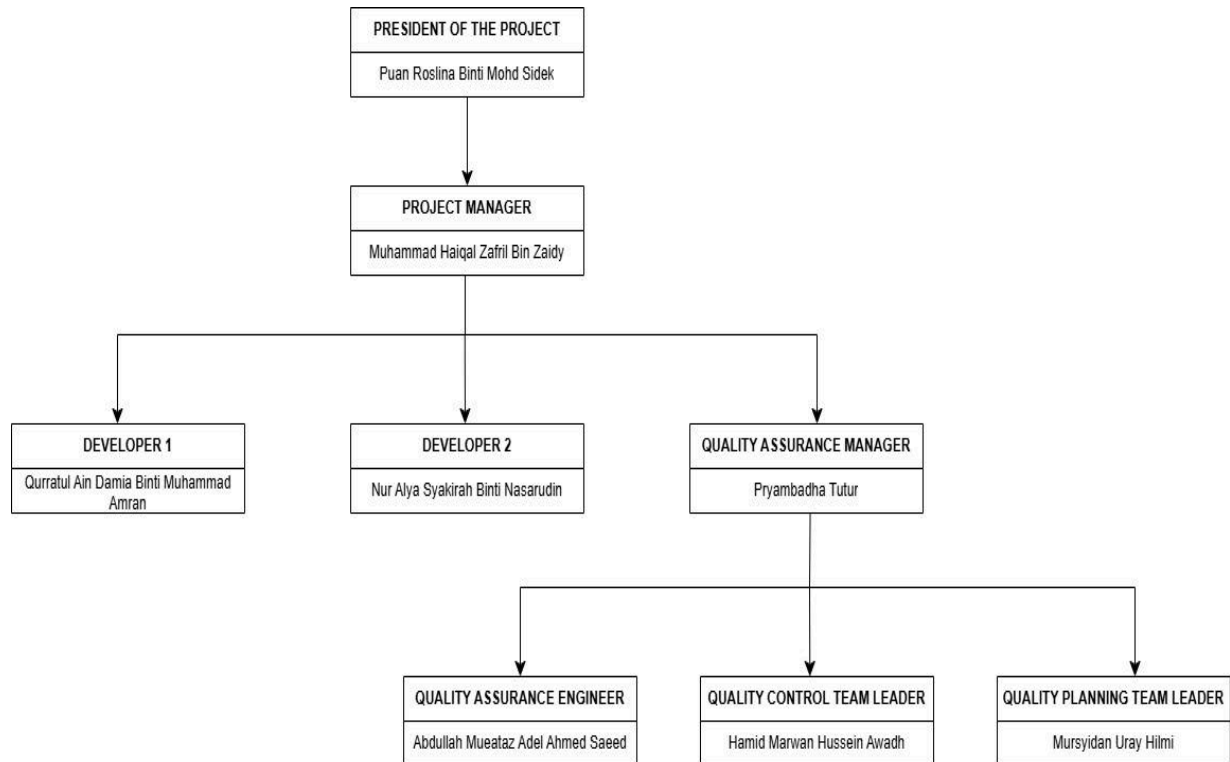
ACRONYM	DEFINITION
IEEE	Institute of Electrical and Electronic Engineers
ISO	International Organization for Standardization
TLMS	Teaching and Learning Management System
SQAP	Software Quality Assurance Plan
SQA	Software Quality Assurance
QA	Quality Assurance
CRUD	Create, Read, Update, Delete
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
PHP	Hypertext Preprocessor
UAT	User Acceptance Test

3.0 REFERENCES DOCUMENTS

- [1] Griffiths, J. (2025, January 20). An introduction to usability testing for software QA. *TestDevLab* Blog. <https://www.testdevlab.com/blog/an-introduction-to-usability-testing-for-software-qa>
- [2] Hazra, S. (2022, September 21). *Software Quality assurance (SQA): definition, plan, and best practices*. Talent500 Blog. <https://talent500.com/blog/what-is-a-sqa-plan/>
- [3] IEEE. (2014, June 13). *IEEE standard for software quality assurance processes (IEEE Std 730-2014)*. IEEE Xplore. <https://ieeexplore.ieee.org/document/6835311>
- [4] International Organization for Standardization. (2017). *ISO/IEC/IEEE 12207:2017 — Systems and software engineering — Software life cycle processes*. ISO. <https://www.iso.org/standard/63712.html>
- [5] K-MOOC Documentation. (n.d.). *Managing course discussions — Building and running an edX course*. https://kmooc.readthedocs.io/en/latest/running_course/discussions.html
- [6] Westland, J. (2025, May 12). *The quality assurance Process: roles, methods & tools*. ProjectManager. <https://www.projectmanager.com/blog/quality-assurance-and-testing>

4.0 SQA PLAN OVERVIEW

4.1 ORGANIZATION AND INDEPENDENCE



ROLE	ASSIGNED TO	RESPONSIBILITY
Project Manager	Muhammad Haiqal Zafril Bin Zaidy	- Provides overall project leadership and coordination. - Plans, schedules, and monitors system development activities. - Oversees module integration and system progress. - Handles the Manage Profile Module.

Developer 1	Qurratul Ain Damia Binti Muhammad Amran	• Designs and implements project solutions based on requirements. • Writes, tests and maintains front-end and back-end code. • Debugs and troubleshoots issues during module development. • Handles the Manage Assignment and Notes Module.
Developer 2	Nur Alya Syakirah Binti Nasarudin	• Develops and implements coding solutions for assigned features. • Writes, tests, and maintains responsive user interfaces and database connections. • Identifies and resolves software defects efficiently. • Handles the Manage Announcement and Discussion Module.
Quality Assurance Manager	Pryambadha Tuter	• Oversees the quality assurance process for the EduQuest

		project. • Defines testing methodologies and quality standards. • Coordinates with QA Engineer and QC Team Leader to ensure compliance. • Ensures the final software product meets functional and performance standards.
Quality Assurance Engineer	Abdullah Mueataz Adel Ahmed Saeed	• Conducts software testing and identifies defects or issues. • Tracks and documents bugs using systematic testing methods. • Performs regression testing to verify fixes. • Evaluates test metrics and suggests improvements in quality assurance processes.
Quality Control Team Leader	Hamid Marwan Hussein Awadh	• Leads a team responsible for inspecting and testing module

		components. • Identifies quality-related defects and reports issues found during inspection. • Collaborates with developers to ensure issues are resolved before deployment. • Maintains detailed quality control documentation and reports.
Quality Planning Team Leader	Mursyidan Uray Hilmi	• Leads the team responsible for developing quality assurance strategies and plans. • Defines quality objectives, key performance metrics, and process guidelines. • Ensures adherence to project standards and promotes continuous improvement. • Coordinates periodic reviews to verify compliance with the SQAP.

4.2 SOFTWARE PRODUCT RISK

RISK	RISK DESCRIPTION	MITIGATION
Data loss	Discussion content or announcements might be accidentally deleted or altered.	Use database backup and validation mechanisms to maintain data integrity and allow recovery.
Inappropriate content	Users may post irrelevant or offensive messages in discussions.	Allow administrators to monitor and remove inappropriate posts to maintain a safe environment.
Broken link or file attachment	Uploaded files or links in announcements may be invalid or inaccessible.	Validate file paths and links during upload to ensure accessibility.
Search inaccuracy	The search feature may not return relevant announcements or discussion threads.	Improve search algorithms and validate keywords for accurate results.

4.3 TOOLS

TOOLS	PURPOSE
Visual Studio Code	Used as the main code editor for HTML, CSS and database integration during development.
XAMPP	Provides a local server environment and database for storing announcements, discussions and user data.
GitHub	Used for version control, project backup and collaborative development among group members.
Draw.io	Utilized to design system architecture diagrams and data flow diagrams.
Microsoft Word	Used to prepare and format documentation, including the project proposal, SQAP and test reports.
Google Drive and Docs	For document storage, sharing and real-time group editing.

4.4 STANDARD, PRACTICES AND CONVENTIONS

STEP OF THE LIFE CYCLE	INTERMEDIATE DELIVERABLE	STANDARD, PRACTICES AND CONVENTIONS
Planning	Project Proposal	PMI - PMBOK® Guide
Planning	Software Quality Assurance Plan (SQAP)	IEEE 730-2014
Design	Interface Mockups	IEEE/ISO/IEC 1016
Development	Source Code	Java programming rules revised April, 20 1999
Development	Coding and Documentation	Visual Studio Code standards and EduQuest coding conventions
Transition to production	Technical Documentation	A local production criteria and checklist

4.5 EFFORT, RESOURCES AND SCHEDULE

4.5.1 EFFORT

- **SQAP DEVELOPMENT**

The Software Quality Assurance Plan establishes quality processes, standards and procedures to ensure that the project meets overall project quality objectives [4]. It defines testing strategies, verification activities, and performance measures to maintain reliability and functionality throughout development.

- **DOCUMENTATION**

All documentation activities will be maintained consistently throughout the project including requirements, design specifications, test cases and validation reports. This ensures clear tracking of module progress and adherence to defined quality standards.

- **MANAGE PROFILE MODULE**

This module enables users to register, update and manage their personal and academic information. It includes functions such as login authentication, profile editing and role management to maintain accurate and secure user data.

- **MANAGE ASSIGNMENT AND NOTES MODULE**

This module allows lecturers to upload, edit and delete learning materials and assignments while students can view, download and submit their work. Administrators can monitor the content to ensure accuracy and quality.

- **MANAGE ANNOUNCEMENT AND DISCUSSION MODULE**

This module focuses on managing all communication features between administrators, lecturers and students. It includes functions for posting, viewing, editing and deleting announcements, as well as creating and replying to discussion threads [5]. The module enhances engagement and information sharing while maintaining appropriate communication standards.

- QUALITY ASSURANCE

Continuous testing, monitoring, and validation processes will be implemented across all stages of development to ensure compliance with the IEEE 730-2014 standard. Quality assurance activities will include requirement validation, usability testing, and defect tracking to verify that the system meets user expectations and institutional goals [6].

4.5.2 RESOURCES

- **HUMAN RESOURCES**

The development team consists of a project manager, two developers, a quality assurance manager, a quality assurance engineer, a quality control team leader and a quality planning team leader . Each member is responsible for specific modules and contributes to integration and testing to ensure system quality.

- **TOOLS AND EQUIPMENT**

The development tools include Visual Studio Code for coding, XAMPP for database management and GitHub for version control and collaboration. Draw.io is used to design data flow diagrams and interface mockups, while Microsoft Word is used for creating and formatting documents such as the SQA Plan and reports.

- **DOCUMENTATION**

All documentation will include the Project Proposal and this Software Quality Assurance Plan (SQAP). These documents outline functional and non-functional requirements, system design details and quality assurance procedures.

- **COMMUNICATION AND COLLABORATION**

Team members communicate daily via WhatsApp for minor updates and use Google Meet for weekly progress discussions and milestone reviews. Shared documents are stored in Google Drive to ensure version consistency and easy access for all members.

4.5.3 SCHEDULE

PHASE	DURATION
SQAP Development	2 weeks
Documentation	Throughout the project
Project Development	6-8 weeks
Quality Assurance	Throughout the project

5.0 ACTIVITIES, OUTCOMES AND TASKS

5.1 PRODUCT ASSURANCE

5.1.1 FUNCTIONALITY

a. Product Assurance

Product assurance for functionality ensures that the Manage Announcement and Discussion Module delivers all intended communication tasks effectively, accurately, and reliably. This involves verifying that lecturers can create and update announcements, students can view and participate in discussions, and administrators can manage or moderate content when necessary. The primary goal is to ensure that all module functions operate reliably according to system specifications while preventing unauthorized access or misuse of features. Through structured validation and verification processes, the module's core CRUD (Create, Read, Update, Delete) operations are continuously tested to confirm that each user role can perform its designated actions correctly without system errors or data inconsistencies.

b. Evaluate Plans for Conformance

During the planning phase, the functional specifications for each feature including posting, viewing, editing and deleting announcements or discussions are clearly documented. Each functionality is reviewed to ensure it aligns with the appropriate user roles, lecturers can create and update announcements or initiate discussion threads, students can view announcements and participate in discussions through replies and administrators have full privileges to monitor or remove content when necessary. This evaluation ensures that planned functionalities conform to system requirements, maintain proper access control and support effective communication across all user levels.

c. Evaluate Product for Conformance

After development, the module undergoes detailed functional testing to verify that all defined operations perform as specified. CRUD functionality is validated through test scenarios that include posting, editing, replying and deleting announcements or discussions. The tests confirm that lecturers can manage content easily, students can

engage in discussions seamlessly and administrators can execute moderation actions without system failure. Each test result is documented and compared with expected outcomes to verify accuracy. Any discrepancies or failures identified during testing are recorded, analyzed and corrected to ensure conformance between the developed product and the planned requirements.

d. Evaluate Plans for Acceptability

Plans are evaluated for their ability to meet user expectations and institutional communication needs. This includes assessing whether the module supports an efficient and straightforward workflow, clear discussion structure and real-time content updates. Accessibility standards are also reviewed to ensure that all users including those with limited technical skills can navigate the module easily. Furthermore, the functional plans are reviewed by both developers and quality assurance personnel to verify that they represent acceptable, practical and user-centered designs that support everyday use in an academic environment.

e. Evaluate Product Life Cycle Support for Conformance

The Manage Announcement and Discussion Module is developed with maintainability and scalability in mind. Its modular design allows for smooth integration of future features such as push notifications or enhanced filtering systems. The system's code structure promotes reusability and simplifies debugging, ensuring that maintenance activities can be performed efficiently. Regression and integration testing are performed regularly to confirm that any modifications or updates do not interfere with existing features or cause unexpected behavior. Through continuous verification, the module maintains stability and functionality throughout its lifecycle.

f. Measure Products

Product functionality is measured through quantitative and qualitative metrics. Quantitative metrics include the number of successful CRUD operations, the rate of defect discovery during testing and the average resolution time for identified issues. Qualitative feedback is obtained through user acceptance testing (UAT), where lecturers, students and administrators interact with the system and provide feedback on functionality and ease of use. These results help assess whether the module performs according to expectations and support future improvements.

g. Outcomes and Tasks

- Ensures all functional requirements operate accurately across user roles.
- Supports future enhancements without compromising existing functions.
- Confirms announcements and discussions perform CRUD operations without error.
- Maintains quality and stability through continuous testing and regression validation.

5.1.2 PERFORMANCE

a. Product Assurance

Product assurance for performance ensures that the Manage Announcement and Discussion Module performs efficiently and remains responsive under different workloads. Since multiple users may access, post or read content simultaneously, the module must maintain optimal speed and reliability. Performance assurance focuses on verifying that database queries execute efficiently and discussions update in real-time. These measures ensure that the system remains practical and responsive during peak academic activity such as examination announcements.

b. Evaluate Plans for Conformance

During the planning stage, performance goals are defined in measurable terms including target response times, system throughput and maximum concurrent users. These benchmarks are reviewed against system specifications to ensure feasibility and conformance to project requirements. The performance plan also includes measures to handle increased load and optimize database queries. Evaluations of these plans ensure that the module's architecture and database structure support scalable performance aligned with institutional needs.

c. Evaluate Product for Conformance

Once implemented, the system is tested under simulated real-world conditions. Load and stress testing tools are used to generate multiple concurrent users accessing announcements or discussions simultaneously. The results are analyzed to measure server response time, database efficiency and data retrieval speed. Identified performance bottlenecks such as slow queries or delayed updates are optimized to improve overall responsiveness. The results confirm whether the product meets defined performance standards and can handle actual classroom activity loads effectively.

d. Evaluate Plans for Acceptability

Performance acceptability is determined by comparing planned performance metrics with user expectations. For example, lecturers and students expect announcement pages to load instantly and discussions to update in real-time without lag. Plans are

reviewed to ensure that these goals are practical and meet user satisfaction standards. The evaluation ensures that planned performance levels align with operational demands maintaining consistency even when user traffic is high.

e. Evaluate Product Life Cycle Support for Conformance

Performance assurance extends throughout the module's lifecycle. Continuous monitoring tools track server performance, load time and database response as the system evolves. Regular load testing after updates ensures that new features do not degrade speed or introduce inefficiencies. Any observed decline in performance triggers optimization measures ensuring that the system maintains its responsiveness and reliability over time.

f. Measure Products

Performance is evaluated using metrics such as average page load time and database query speed. These measurements are recorded during testing and after deployment to identify performance trends and ensure long-term stability. Performance benchmarks are periodically reviewed to align with system upgrades and new user requirements.

g. Outcomes and Tasks

- Ensures stable and responsive system performance across all usage levels.
- Prevents system lag through optimization and load testing.
- Confirms continued performance through lifecycle monitoring and testing.
- Enhances user experience by maintaining quick, reliable access to communication tools.

5.1.3 USABILITY

a. Product Assurance

Product assurance for usability ensures that the Manage Announcement and Discussion Module is intuitive, accessible and easy to navigate for all users. The module's interface is designed to simplify communication, allowing users to locate announcements and participate in discussions. This assurance focuses on providing a consistent and visually clear layout that supports the natural flow of user actions, reducing the likelihood of errors and confusion.

b. Evaluate Plans for Conformance

The design plans for the module are reviewed during the planning stage to ensure compliance with usability standards. This includes ensuring readable typography, clear button placement, logical navigation and responsive design. Accessibility guidelines, such as proper color contrast and text clarity are also followed. These design evaluations confirm that the system's visual and functional elements are easy to use and understand, even for first-time users.

c. Evaluate Product for Conformance

Usability testing is performed by observing actual users, lecturers, students and administrators interacting with the system. Tasks such as posting announcements, viewing discussions and replying to threads are tested to identify points of confusion or difficulty. Feedback is collected through surveys and interviews allowing developers to make interface adjustments that improve overall user experience. This process ensures that the final product conforms to user needs and expectations.

d. Evaluate Plans for Acceptability

Usability plans are evaluated to confirm that they address the practical needs of all system users. Lecturers must be able to manage content efficiently, students must navigate and respond easily and administrators must moderate without complexity. Plans are reviewed to ensure that all user interfaces promote efficiency and are accessible to a wide range of users with varying technical skills. Acceptability evaluations confirm that the interface design enhances communication and engagement.

e. Evaluate Product Life Cycle Support for Conformance

As the system evolves, usability must remain consistent. Updates to the user interface undergo regression testing to ensure new features do not disrupt the familiar layout or introduce unnecessary complexity. Documentation, tutorials and guidance are updated alongside interface changes to maintain high usability standards throughout the module's lifecycle.

f. Measure Products

Usability is measured through task success rate, error frequency, time to complete key actions and user satisfaction ratings. Post-deployment feedback and periodic surveys provide valuable data to evaluate user experience and identify improvement opportunities.

g. Outcomes and Tasks

- Ensures the interface remains user-friendly and visually consistent.
- Maintains usability quality through ongoing feedback and testing.
- Supports positive user experience across all roles in the EduQuest system.
- Enhances accessibility and reduces user frustration through design optimization.

5.2 PROCESS ASSURANCE

a. Evaluate Life Cycle Processes for Conformance

All development processes including requirement analysis, design, coding, testing, deployment and maintenance are reviewed for conformance with IEEE 730 standards. Each stage includes quality checkpoints to verify compliance with project requirements and ensure effective documentation. Peer reviews, change tracking and testing records are maintained to guarantee consistency and accountability across the software lifecycle.

b. Evaluate Environments for Conformance

The testing and deployment environments are evaluated to ensure they replicate real-world usage conditions including database performance, user load and internet connectivity. Properly configured environments help validate that the module performs consistently once deployed. This evaluation reduces the likelihood of environment-related defects after release.

c. Evaluate Subcontractor Processes for Conformance

Third-party tools and external contributors such as hosting services, interface libraries or testing utilities, are required to comply with the project's established quality standards and procedures. Their development and testing processes are evaluated to ensure alignment with the team's defined performance, security and integration criteria. This assessment guarantees that all externally sourced components operate reliably within the system environment, maintaining overall stability, compatibility, and data integrity without introducing additional risks to the Manage Announcement and Discussion Module.

d. Measure Processes

The project's process quality is tracked using metrics such as defect density, issue resolution time and test coverage percentage. Continuous measurement provides insight into process efficiency, identifies bottlenecks and supports ongoing improvement. Reports generated from these metrics are reviewed in QA meetings for action planning.

e. Assess Staff Skill and Knowledge

The development and QA teams' technical proficiency in HTML, CSS, MySQL and quality assurance methodologies is periodically assessed. Training sessions and collaborative workshops are conducted to strengthen knowledge in testing techniques, debugging and secure coding practices. Continuous learning ensures that all team members are capable of maintaining and improving the quality of the Manage Announcement and Discussion Module throughout its lifecycle.

6.0 ADDITIONAL CONSIDERATIONS

6.1 CONTRACT REVIEW

Contract review represents a formal evaluation of all project agreements, requirements and responsibilities related to the Manage Announcement and Discussion Module. This process ensures that every stakeholder clearly understands their roles, deliverables and expectations regarding quality assurance and development. The review confirms that the established scope, schedule and quality standards are aligned with institutional goals and system objectives. Through this process, any ambiguities, conflicting requirements or unrealistic expectations are identified and resolved prior to implementation. A well-conducted contract review guarantees that all development and testing activities proceed under clear and mutually agreed conditions reducing the risk of miscommunication or non-compliance.

6.2 QUALITY MEASUREMENT

Quality measurement serves as a continuous process to evaluate the performance, reliability and usability of the Manage Announcement and Discussion Module. Specific quality indicators such as response time, database consistency, user satisfaction rate and system uptime are systematically measured throughout the development lifecycle. The QA team records data from testing and user evaluations to determine whether established benchmarks are achieved. These measurements enable early detection of quality gaps and guide the development team in implementing corrective actions. Consistent monitoring and evaluation ensure that the module fulfills defined quality objectives and maintains high standards of functionality and performance.

6.3 WAIVERS AND DEVIATIONS

Waivers and deviations refer to formally approved exceptions from the initially defined requirements or standards within the project. Such cases arise when technical limitations, time constraints or resource availability prevent full adherence to original specifications. Each waiver is carefully reviewed, documented and evaluated by the project manager and QA manager to assess potential impacts on overall quality and user satisfaction. For example, advanced features such as threaded replies or multimedia attachments may be deferred to future updates if not feasible in the current phase. The approval and documentation of waivers ensure transparency, accountability and preservation of software integrity.

6.4 TASK REPETITION

Task repetition constitutes a necessary part of maintaining software quality throughout the project lifecycle. Activities such as regression testing, usability evaluation and documentation review are repeated after each modification or update to verify that existing features continue to function correctly. The repetition of testing tasks ensures that the module's features such as posting announcements, replying to discussions and content moderation remain stable and reliable after system updates. This repetitive process enhances the consistency of results, reduces the likelihood of recurring defects and confirms that the system continues to meet its intended quality standards over time.

6.5 RISK TO PERFORMING SQA

Risk to perform Software Quality Assurance (SQA) encompasses potential challenges that may affect the effectiveness of testing, monitoring or validation activities. Common risks include limited manpower, inadequate testing tools, incomplete requirements and communication barriers between development and QA teams. These factors may result in delayed testing, missed defects or reduced quality coverage. To mitigate such risks, the team enforces proper scheduling, ensures complete documentation of requirements and conducts periodic review meetings. Continuous coordination between the QA manager, developers and

stakeholders strengthens project oversight and ensures that software quality assurance objectives are successfully achieved.

6.6 COMMUNICATIONS STRATEGY

Communication strategy defines the structured approach used to facilitate information flow and coordination throughout the project. All team members maintain clear and consistent communication channels using WhatsApp for daily task updates and Google Meet for formal project discussions and milestone reviews. Project documentation, including SQAP reports, test plans and meeting minutes is stored in Google Drive for easy access and version control. Regular progress meetings allow members to discuss challenges, share findings and propose improvements. This structured communication process ensures transparency, promotes accountability and supports timely decision-making across all project activities.

6.7 NON-CONFORMANCE PROCESS

Non-conformance process establishes a systematic method for identifying, documenting, analyzing and resolving issues that deviate from the expected software quality standards. Any discrepancies detected through testing, audits or user feedback are recorded and categorized according to their severity and impact. Root cause analysis is performed to determine underlying issues, followed by corrective and preventive actions verified through retesting. All non-conformance records are documented and reviewed by the QA manager for traceability and process improvement. Implementing this structured approach ensures that the Manage Announcement and Discussion Module maintains reliability and continuous compliance with defined quality objectives.

7.0 SQA RECORDS

7.1 ANALYZE, IDENTIFY, COLLECT, FILE, MAINTAIN AND DISPOSE

The Software Quality Assurance records for the Manage Announcement and Discussion Module are maintained to ensure traceability, accuracy and consistency throughout the project. All key documents such as requirements, test plans, test reports, defect logs and review summaries are identified, collected and properly stored.

The QA Manager is responsible for organizing and maintaining these records in shared repositories like Google Drive and GitHub. Regular reviews are conducted to ensure the records remain relevant and up to date. Outdated documents are archived or securely disposed of to prevent duplication and maintain an organized record system.

This structured management of records ensures proper documentation of all quality assurance activities and supports effective auditing and continuous improvement.

7.2 AVAILABILITY OF RECORDS

SQA records are made accessible to authorized team members including the Project Manager, Developers and Quality Assurance personnel. The QA Manager regulates access to maintain confidentiality and prevent unauthorized alterations.

Digital copies are stored in Google Drive for convenient retrieval and backed up regularly to ensure data safety. During review sessions or audits, required documents such as testing reports and validation records can be quickly retrieved. Upon project completion, finalized documents are compiled and submitted for official archiving and future reference.

This process guarantees that project documentation remains secure, well-organized and readily available for quality verification and evaluation.